

1. Image formation (2)
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4. 考点分析与习题演练

B. 课时 2

1. Image features (6)
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D. 课时 4

1. Object detection and segmentation (1)
2. Motion estimation & Object Tracking (2)
3. 考点分析与习题演练

Q-卷积分离

Question: Suppose we have a 2D image that is very noisy and we want to smooth it by filtering with a uniform kernel of size 5×5 voxels. Let S be the total number of floating-point operations required per pixel to perform the operation when using a dimensionally separated implementation of the filter. And let N be the total number of floating-point operations required per pixel to perform the operation when using a dimensionally nonseparated implementation of the filter.

What is the ratio S/N (as a percentage rounded to one decimal)?

- A. 36.7%
B. 40.0%
C. 58.8%
D. 66.7%

这是一道关于图像处理中**滤波 (Filtering)** 计算复杂度的经典题目。我们需要比较二维滤波器 (2D Filter) 在**不可分离 (Non-separated)** 实现和**可分离 (Separated)** 实现下的浮点运算次数 (FLOPs)。

核心概念

- 卷积核大小 (Kernel Size)**：题目给定 $k = 5$ 。
- 基本运算**：对于每一个像素的卷积操作，通常包含乘法 (Multiplications) 和加法 (Additions)。

第一步：计算不可分离 (Non-separated) 实现的运算量 N

直接使用 $k \times k$ 的二维核进行卷积。

对于图像上的每一个像素，我们需要将核与对应的图像区域进行点积：

- 乘法次数**： $k \times k$ 次 (每个权重对应一个像素值)。
- 加法次数**： $k \times k - 1$ 次 (将所有乘积结果相加)。

总运算量 N 为：

$$N = k^2 + (k^2 - 1) = 2k^2 - 1$$

代入 $k = 5$ ：

$$N = 2 \times 5^2 - 1 = 50 - 1 = 49 \text{ (FLOPs per pixel)}$$

第二步：计算可分离 (Dimensionally Separated) 实现的运算量 S

一个二维均匀滤波器 (Uniform Filter) 是可分离的，它可以分解为两个一维滤波器 (一个 $1 \times k$ 的行滤波器和一个 $k \times 1$ 的列滤波器)。

这种方法分两步进行：

1. **第一步 (行方向)**：使用 $1 \times k$ 核对图像进行滤波。
 - 运算量： k 次乘法 + $(k - 1)$ 次加法 = $2k - 1$ 。
2. **第二步 (列方向)**：使用 $k \times 1$ 核对第一步的结果进行滤波。
 - 运算量： k 次乘法 + $(k - 1)$ 次加法 = $2k - 1$ 。

总运算量 S 为：

$$S = (2k - 1) + (2k - 1) = 2(2k - 1)$$

代入 $k = 5$ ：

$$S = 2 \times (2 \times 5 - 1) = 2 \times 9 = 18 \text{ (FLOPs per pixel)}$$

第三步：计算比率 S/N

$$\frac{S}{N} = \frac{18}{49}$$

进行除法运算：

$$18 \div 49 \approx 0.36734\dots$$

转换为百分比并保留一位小数：

$$\approx 36.7\%$$

公式

可以写成一般的公式，你记住一个 $k \times k$ 卷积核的通式就行。

设卷积核大小为 $k \times k$ (本题 $k = 5$)：

1 非分离 2D 卷积 (N)

- 乘法次数: k^2
- 加法次数: $k^2 - 1$

每个像素要把 k^2 个数相乘, 然后把它们加起来。

所以总浮点运算数 (乘 + 加) :

$$N = k^2 + (k^2 - 1) = 2k^2 - 1$$

2 分离卷积 (先做横向 1D, 再做纵向 1D) (S)

一个长度为 k 的 1D 卷积:

- 乘法次数: k
- 加法次数: $k - 1$

总共 $k + (k - 1) = 2k - 1$ 次浮点运算。

因为要做两次 (一维横向 + 一维纵向) :

$$S = 2(2k - 1) = 4k - 2$$

3 比值公式 S/N

$$\frac{S}{N} = \frac{4k - 2}{2k^2 - 1}$$

代入本题 $k = 5$:

- $N = 2 \cdot 5^2 - 1 = 49$
- $S = 4 \cdot 5 - 2 = 18$

$$\frac{S}{N} = \frac{18}{49} \approx 0.367 \Rightarrow 36.7\%$$

Question 1-D

Which one of the following colour spaces is most perceptually uniform?

- A. The RGB colour space
 - B. The HSV colour space
 - C. The YCbCr colour space
 - D. The Lab* colour space ✓ (正确答案)
-

视觉上均匀

Question 2-B

Given a microscope imaging experiment where the number of photons per pixel coming in from the biological sample is expected to be up to 4,000.

What is the minimum number of bits per pixel required to record the photon count without loss of information?

- A. 8
- B. 12 ✓ (正确答案)
- C. 16
- D. 24

★ (附) Question 2 计算解释 (你需要的话可写进正式作业)

最大光子数 = 4000

需要的最小位数:

$$2^n \geq 4000$$

因为:

- $2^{11} = 2048 < 4000$
- $2^{12} = 4096 \geq 4000$

所以:

$$n = 12 \text{ bits}$$

解析

比特数与可表示的最大数值的关系是: n 比特可表示的最大整数为 $2^n - 1$ (从 0 开始计数)。

需要找到最小的 n , 使得 $2^n - 1 \geq 4000$:

- $2^{11} - 1 = 2047$ (小于 4000)
- $2^{12} - 1 = 4095$ (大于 4000)

因此, 至少需要 12 比特才能完整记录最多 4000 的光子数。

答案

正确选项是 **B**。

Q3-C

Question 3

Which one of the following statements about intensity transformations is incorrect?

- A. Contrast stretching linearly maps intensities between two values to the full output range.
 - B. Log transformation maps a range of very low intensities to a wider range of output values.
 - C. Power transformation represents a family of transformations including log transformation.
 - D. Piecewise linear transformations can achieve contrast stretching and intensity slicing.
-

逐条判断：

- **A**：对比度拉伸就是在输入的两个灰度值之间做**线性映射**，把它们拉到整个输出范围上——这是对的。
- **B**：对数变换把一小段**低灰度**拉成更宽的输出范围，用来增强暗部细节——也是正确描述。
- **C**：幂次 (γ) 变换 $s = cr^\gamma$ 是一族“power-law”变换；**对数变换** $s = c \log(1 + r)$ **并不是其中一个幂次特例**，是另一种单独的函数族。所以说“包括 log 变换”是错误的。
- **D**：分段线性变换可以做“对比度拉伸”和“强度切片 (intensity slicing)”，这是书上经典例子，也是对的。

所以，**不正确的说法是 C。**

Q4-直方图均衡化-C

Question 4

Which one of the following statements about histogram equalization is incorrect?

- A. The mapping function for histogram equalization is calculated from the cumulative histogram.*
 - B. Applying a single mapping function to the whole image does not yield a truly equalized histogram.*
 - C. Histogram equalization increases contrast for intensity values near the histogram minima.*
 - D. In practice the slope of the mapping function for histogram equalization is often constrained.*
-

直译一下意思：

问的是“下面关于**直方图均衡化**的说法哪一个是**错误的**？”

- A：映射函数是由累积分布（累积直方图）算出来的。✓
- B：对整幅图只用一个映射函数，一般得不到严格均匀的直方图。✓（因为灰度是离散的等原因）
- C：直方图均衡化会在直方图的“谷底附近”的灰度上增强对比度。✗
- D：在实际应用中，映射函数的斜率往往会被限制（防止过度放大噪声等）。✓

enhance maxima, decrease minima

Q5-B

Question 5

Suppose an image analysis software tool has a command named “Segment > Threshold” which can automatically segment images by thresholding. It is not clear from this naming what thresholding method is actually used. In the manual of the software tool, the command is described as follows: “It iteratively computes the mean intensity of the pixels above (inclusive) and below an initial threshold and takes the mean of these means as the new threshold until convergence.” What is the actual name of this thresholding method?

- A. Mean thresholding
 - B. IsoData thresholding
 - C. Otsu thresholding
 - D. Triangle thresholding
-

这段描述说的是：

- 先选一个初始阈值 T
- 计算大于等于 T 的像素的平均灰度 m_1
- 计算小于 T 的像素的平均灰度 m_2
- 新阈值 $T' = (m_1 + m_2)/2$
- 不断迭代直到收敛

这是典型的 **IsoData / iterative intermeans thresholding** 方法，而不是 Otsu、Triangle 或简单“均值一次阈值”。

👉 正确答案：B. IsoData thresholding



IsoDat

segment mean

Q6-C-频域空域

Which one of the following statements about frequency-domain processing is incorrect?

- A. Any digital image can be written as the sum of a finite number of frequencies.
 - B. Noise suppression in images can be obtained by removing high-frequency components.
 - C. Convolution in the frequency domain corresponds to correlation in the spatial domain.
 - D. Changing one point in the frequency domain may affect all pixels in the spatial domain.
-

这是一道关于**频域图像处理**的选择题，需逐一分析选项的正确性：

选项分析

1. **选项 A**数字图像是离散信号，根据离散傅里叶变换（DFT），其频域分量是**有限个频率的组合**（因为数字图像尺寸有限）。**该表述正确。**
2. **选项 B**图像中的噪声通常对应**高频分量**（细节 / 突变信息），因此在频域中去除高频分量可实现噪声抑制。**该表述正确。**
3. **选项 C**根据傅里叶变换的性质：**空间域的卷积对应频域的乘积**；而**空间域的相关对应频域的共轭乘积**。二者并非“频域卷积对应空域相关”的关系。**该表述错误。**
4. **选项 D**频域与空域是全局变换（傅里叶变换是全局操作），因此频域中单个点的修改会影响空域所有像素。**该表述正确。**

答案

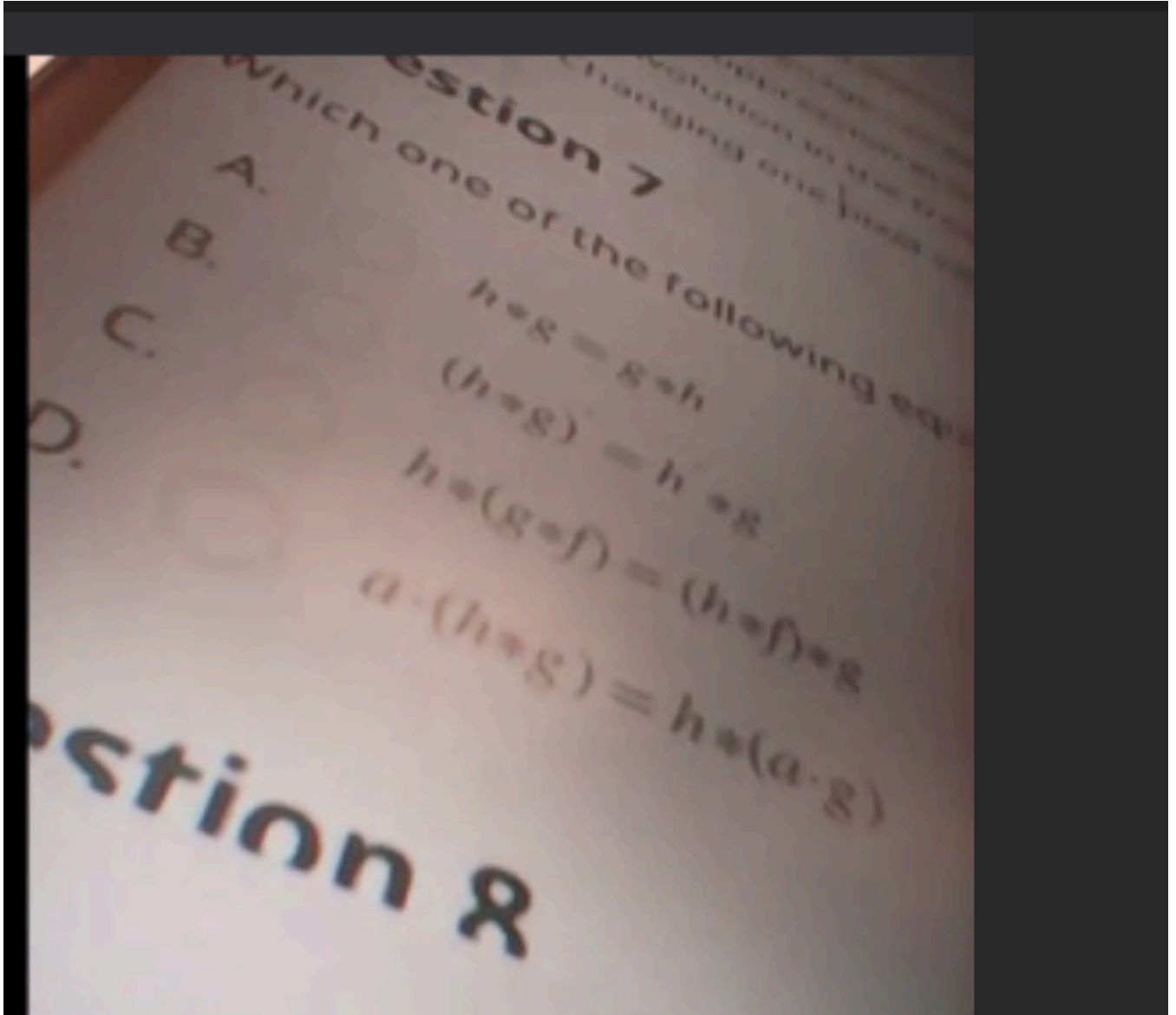
错误的选项是 *C*。

B. Properties of the Fourier transform

Property	Spatial	Frequency
Superposition	$f_1(x) + f_2(x)$	$F_1(u) + F_2(u)$
Translation	$f(x - \Delta x)$	$F(u)e^{-i2\pi u \Delta x}$
Convolution	$f(x) * h(x)$	$F(u)H(u)$
Correlation	$f(x) \otimes h(x)$	$F(u)H^*(u)$
Multiplication	$f(x)h(x)$	$F(u) * H(u)$
Scaling	$f(ax)$	$F(u/a)/a$
Differentiation	$f^{(n)}(x)$	$(i2\pi u)^n F(u)$

- 空域：叠加 ↔ 频域：相位调整
- 空域：平移 ↔ 频域：幅度调整
- 空域：卷积 ↔ 频域：乘积
- 空域：相关 ↔ 频域：共轭乘积
- 空域：乘积 ↔ 频域：卷积
- 空域：缩放 ↔ 频域：反向缩放
- 空域：微分 ↔ 频域：多项式加权

Q7-C-卷积



Question 7

Which one of the following operations concerning convolution (*) of images (h, g, f) is incorrect?

- A. $h * g = g * h$
- B. $(h * g)' = h' * g$
- C. $h * (g * f) = (h * f) * g$
- D. $a(h * g) = h * (ag)$

卷积运算的基本性质分析：

1. **选项 A：**卷积满足**交换律**，即 $h * g = g * h$ 。**表述正确。**
2. **选项 B：**卷积的导数性质应为 $(h * g)' = h' * g = h * g'$ （卷积与微分可交换），该表述符合规则。**表述正确。**
3. **选项 C：**卷积满足**结合律**，但结合律的形式是 $h * (g * f) = (h * g) * f$ （而非 $(h * f) * g$ ）。此选项混淆了结合律与交换律，**表述错误。**
4. **选项 D：**卷积满足**数乘结合律**，即 $a(h * g) = h * (ag) = (ah) * g$ （a 为常数）。**表述正确。**

答案

错误选项是 C 。

Q8-A-一维卷积

Suppressing or enhancing specific image features is often done by convolution with a kernel. For example, to approximately compute the second-order derivative in 1D, the following kernel can be applied:

$$\begin{bmatrix} 1 & -2 & 1 \end{bmatrix}$$

One way to make an approximation of the fourth-order derivative is to apply this kernel twice:

first to the input image and then to the resulting image.

Which one of the following kernels has the same effect as applying the above kernel twice?

二阶导数核 / Second-order derivative kernel: $[1, -2, 1]$

问题: 哪个卷积核与应用上述核两次的效果相同?

Which kernel has the same effect as applying the above kernel twice?

选项 / Options:

- A) $[1, -4, 6, -4, 1]$
- B) $[2, -4, 2]$
- C) $[-2, 4, -2]$
- D) $[1, 0, -2, 0, 1]$

选项是四个一维卷积核，其中一个是长度 5 的:

$$\begin{bmatrix} 1 & -4 & 6 & -4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -4 & 2 \end{bmatrix}$$

$$\begin{bmatrix} -2 & 4 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -2 & 0 & 1 \end{bmatrix}$$

计算

把核 $k = [1, -2, 1]$ 和自己做一次离散卷积:

$$\begin{aligned} h &= k * k \\ &= [1, -2, 1] * [1, -2, 1] \\ &= [1, -4, 6, -4, 1] \end{aligned}$$

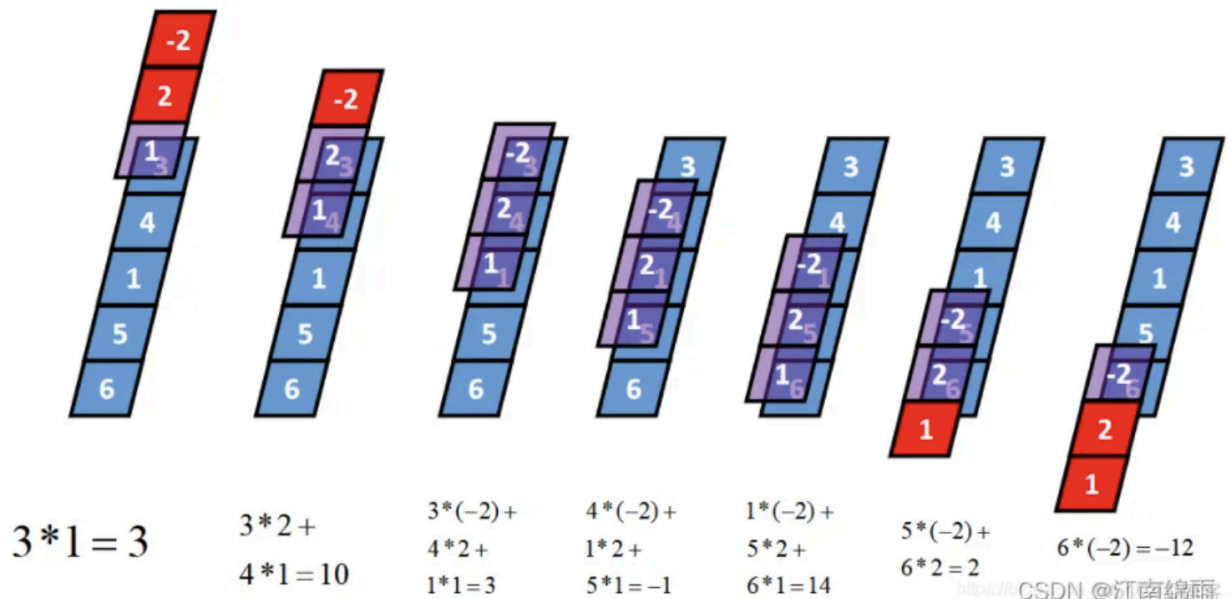
所以等价的单次卷积核是:

$$[1, -4, 6, -4, 1]$$

👉 选择对应这个核的选项（那一条写成 1 4 6 4 1 的那一行，就是 A）。

I：一维Full卷积

Full卷积的计算过程是：K沿着I顺序移动，每移动到一个固定位置，对应位置的值相乘再求和，计算过程如下：



4. 核与核的卷积 / Kernel Convolution

题目中的计算: $[1, -2, 1] * [1, -2, 1]$

方法: 滑动并计算

步骤1: 对齐最左边

$[1, -2, 1]$
 $[1, -2, 1]$
↑
只有 $1 \times 1 = 1$

步骤2: 滑动一位

$[1, -2, 1]$
 $[1, -2, 1]$
↑ ↑
 $1 \times (-2) + (-2) \times 1 = -2 - 2 = -4$

步骤3: 完全重叠

$[1, -2, 1]$
 $[1, -2, 1]$
↑ ↑ ↑
 $1 \times 1 + (-2) \times (-2) + 1 \times 1 = 1 + 4 + 1 = 6$

步骤4: 继续滑动

$[1, -2, 1]$
 $[1, -2, 1]$
↑ ↑
 $(-2) \times 1 + 1 \times (-2) = -2 - 2 = -4$

步骤5: 最右边

$[1, -2, 1]$
 $[1, -2, 1]$
 ↑
 $1 \times 1 = 1$

结果: $[1, -4, 6, -4, 1]$

参考

https://blog.51cto.com/u_16213565/10256662

Q9-D

Question 9

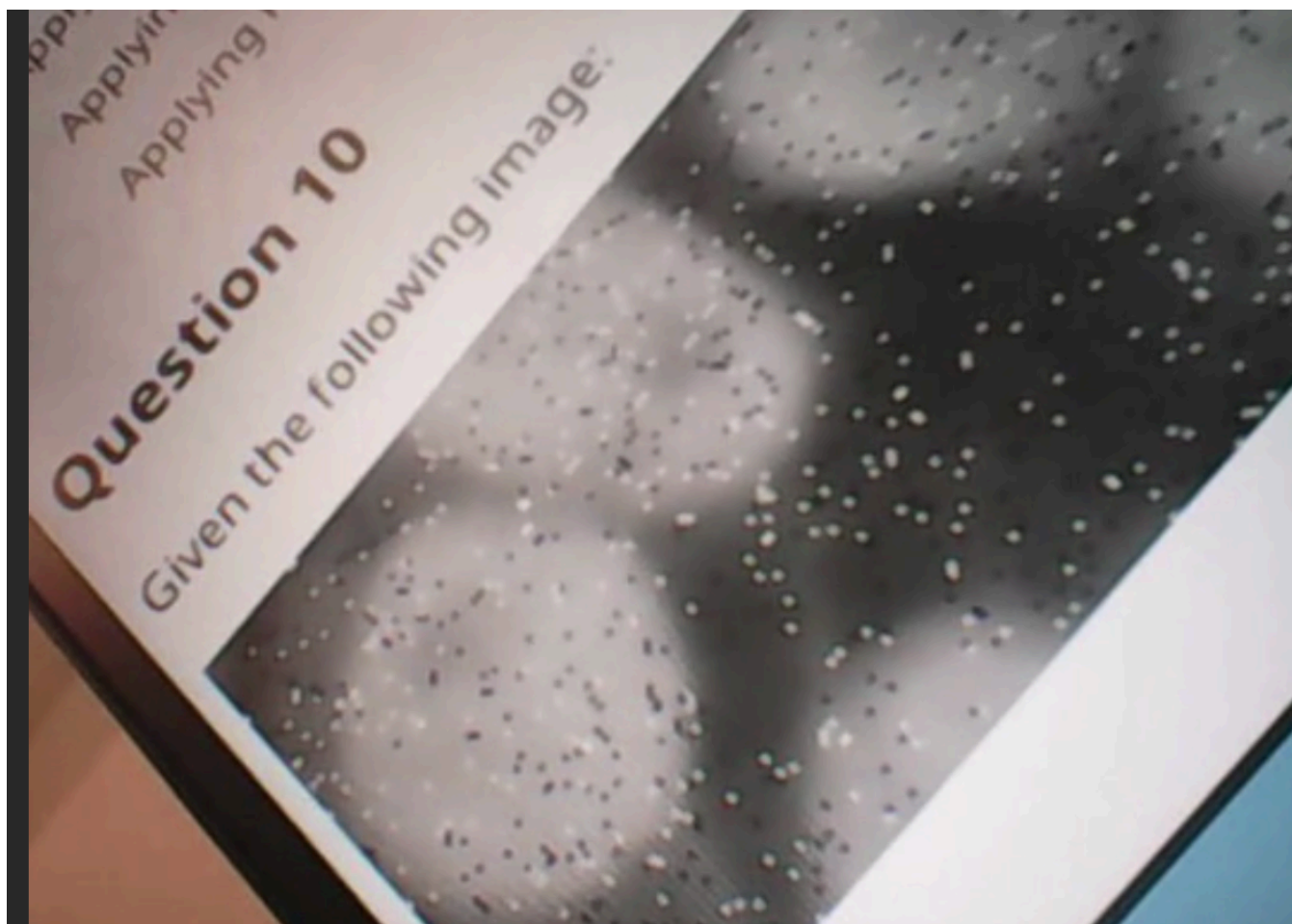
Which one of the following image processing operations does **not** result in noise reduction?

- A. Applying median filtering with a kernel of radius 3 pixels.
- B. Applying averaging of multiple images of the same scene.
- C. Applying local averaging with a uniform 3×3 kernel.
- D. Applying histogram equalization to the image.

解析

- 中值滤波 (Median Filtering) → 去除椒盐噪声效果好 → **减噪**
- 多帧平均 (Averaging multiple images) → 随机噪声相互抵消 → **减噪**
- 3×3 均值滤波 (Local averaging) → 平滑噪声 → **减噪**
- 直方图均衡化 (Histogram Equalization) → 增强对比度
 - 可能会放大噪声，并不会减少噪声 → **不属于降噪操作**

Q10-B



Question 10 (完整复原)

Given the following image:

(图像展示大量黑白噪点 — 椒盐噪声)

The image has many corrupted pixels (the black and white spots). Which one of the following image filtering operations best removes these pixels while keeping the rest of the image intact as much as possible?

- A. Uniform filtering
- B. Median filtering
- C. Gaussian filtering
- D. Laplacean filtering

✔ 正确答案: B. Median filtering

🔍 解析 (简洁版)

方法	是否适合去除椒盐噪声	是否模糊边缘
Uniform (均值滤波)	部分去除但模糊较明显	边缘模糊
Gaussian (高斯滤波)	对随机噪声有效, 但对椒盐噪声不佳	会模糊
Laplace (拉普拉斯)	边缘增强, 会放大噪声	否
Median (中值滤波)	效果最佳	保留边缘

Q11-C-高斯滤波

Question 11

Which one of the following statements about the continuous Gaussian filter is incorrect?

- A. The Gaussian filter is the only filter that is both separable and circularly symmetric.
- B. The Fourier transform of the Gaussian filter is a Gaussian filter in the frequency domain.
- C. The n-fold convolution of any high-pass filter converges to a Gaussian filter.
- D. The derivative of the Gaussian filter can always be calculated to any desired degree.

Based on the image provided, here is the bilingual explanation.

The correct answer is C.

正确答案是 C。

Detailed Explanation (详细解释)

- **C. The n-fold convolution of any high-pass filter converges to a Gaussian filter. (INCORRECT)**
 - **English:** This statement is incorrect because it misapplies the Central Limit Theorem. The Central Limit Theorem suggests that the repeated convolution of a **low-pass filter** (specifically one that is non-negative and has a finite area, like a box filter) converges to a Gaussian. A high-pass filter (which emphasizes edges and usually sums to zero) does not converge to a Gaussian upon repeated convolution.
 - **中文:** 这个说法是不正确的，因为它误用了中心极限定理。中心极限定理表明，**低通滤波器**（特别是那些非负且积分有限的滤波器，如盒状滤波器/Box Filter）的多次重复卷积会收敛于高斯函数。高通滤波器（通常用于增强边缘且总和为零）在重复卷积后并不会收敛于高斯函数。

Why the other options are correct (其他选项为何正确)

- **A. The Gaussian filter is the only filter that is both separable and circularly symmetric.**
 - **English:** True. This is a unique property of the Gaussian. Being separable allows efficient computation (splitting 2D convolution into two 1D convolutions), and being circularly symmetric ensures it smooths the image equally in all directions without directional bias.
 - **中文:** 正确。这是高斯函数的独特属性。可分离性使得计算非常高效（将2D卷积拆分为两个1D卷积），而圆对称性保证了它在所有方向上对图像的平滑效果是一致的，没有方向偏差。
- **B. The Fourier transform of the Gaussian filter is a Gaussian filter in the frequency domain.**
 - **English:** True. The Gaussian is a unique function where its Fourier transform is also a Gaussian. A narrow Gaussian in the spatial domain becomes a wide Gaussian in the frequency domain, but the functional form remains the same.
 - **中文:** 正确。高斯函数是一个独特的函数，其傅里叶变换仍然是高斯函数。空域中较窄的高斯函数在频域中会变成较宽的高斯函数，但其函数形式保持不变。
- **D. The derivative of the Gaussian filter can always be calculated to any desired degree.**
 - **English:** True. The Gaussian function (e^{-x^2}) is a C^∞ smooth function, meaning it is infinitely differentiable. This property is crucial for algorithms like Canny edge detection (Derivative of Gaussian).
 - **中文:** 正确。高斯函数 (e^{-x^2}) 是 C^∞ 光滑函数，意味着它是无限可微的。这一特性对于像 Canny 边缘检测（高斯导数）这样的算法至关重要。

Q12-均值滤波-D

Question 12

给定输入图像 I 和一个邻域 N ，利用下面的操作得到输出图像 O ：

$$O(x, y) = \frac{1}{|N|} \sum_{(i, j) \in N} I(x + i, y + j)$$

Which one of the following statements about this operation is incorrect?

- A. The operation is equivalent to uniform filtering.
- B. The operation is linear and shift-invariant.
- C. The operation reduces background noise.
- D. The operation enhances object edges.

分析：这是对邻域 N 做**平均（均值）滤波**，即均匀滤波。

- A：确实就是 uniform (mean) filtering ✓
- B：卷积 + 常数系数，是线性且平移不变 ✓
- C：均值滤波可以一定程度上平滑/降低随机噪声 ✓
- D：均值滤波会**模糊边缘**，不会增强边缘 ✗

所以不正确的是：

👉 D. The operation enhances object edges.

Q13-D

Question 13

Given a grayscale input image A to which we apply the following algorithm:

- 1. Compute image B by convolving A with a Gaussian filter.
- 2. Compute image C by subtracting B from A pixelwise.
- 3. Compute image D by adding C to A pixelwise.

What is the output image D ?

- A. The same as the input image.
- B. The smoothed version of the input image.
- C. The high-pass filtered version of the input image.
- D. The sharpened version of the input image.

✔ 正确答案: D. The sharpened version of the input image.

🔍 解析

步骤回顾:

Step	Operation	Meaning	🔄
$B = A * G$	Gaussian blur	低通滤波, 平滑	
$C = A - B$	Input - blur	高通信息 (细节/边缘)	
$D = A + C = A + (A - B)$	强化细节	$D = 2A - B$ (Unsharp Masking)	

这个过程就是典型的:

Unsharp Masking (图像锐化)

增强细节和边缘, 因此输出是:

★ Sharpened image

Q14-D

Question 14

Suppose we have a 3D image that is very noisy and we want to smooth it by filtering with a uniform kernel of size $7 \times 7 \times 7$ voxels.

Let **S** be the total number of floating-point operations required per pixel to perform the operation when using a **dimensionally separated** implementation of the filter.

Let **N** be the total number of floating-point operations required per pixel to perform the operation when using a **dimensionally non-separated** implementation of the filter.

What is the ratio S/N (as a percentage rounded to one decimal)?

Options:

- A. 36.7%
- B. 26.8%
- C. 10.8%
- D. 5.7%

解题思路

假设一次加法或一次乘法都算 1 个 floating-point operation (flop)。

1. 非分离 3D 卷积 (non-separated)

核大小: $7 \times 7 \times 7 = 343$ 个系数。

每个输出体素需要:

- 343 次乘法
- 342 次加法 (把 343 个数加起来)

总 flop 数:

$$N = 343 + 342 = 685$$

2. 维度可分 3D 卷积 (separated)

分成 3 个 1D 卷积 (x、y、z 三个方向)，每个方向的 1D 核长度为 7。

对一次 1D 卷积 (长度 7)：

- 7 次乘法
- 6 次加法

共有 3 个方向，所以：

$$S = 3 \times (7 + 6) = 3 \times 13 = 39$$

3. 计算比例 S/N

$$\frac{S}{N} = \frac{39}{685}$$

先估算成百分比：

- $685 \times 0.05 = 34.25$
- 剩余差值： $39 - 34.25 = 4.75$
- $685 \times 0.006 = 4.11$
- $685 \times 0.0007 \approx 0.48$

所以：

$$0.05 + 0.006 + 0.0007 \approx 0.0567 \approx 5.7\%$$

四舍五入到一位小数约为 **5.7%**。

Q15-edge-C

Question 15

Suppose you are given a 2D image I and you are asked to implement an algorithm that computes the edge image $\|\nabla I\|$. On some website you find the following four-step algorithm that supposedly accomplishes that operation. **Which step is incorrect?**

- A. Convolve with a first-order derivative kernel in x and in y .
- B. Square the resulting images from the previous step pixelwise.
- C. Multiply the resulting images from the previous step pixelwise.
- D. Square-root the resulting image from the previous step pixelwise.

计算梯度幅值的正确公式是

$$\|\nabla I\| = \sqrt{I_x^2 + I_y^2}$$

正确步骤应该是：

1. 用一阶导数核在 x 和 y 方向卷积，得到 I_x, I_y ✓ (A)
2. 对 I_x, I_y 各自逐像素平方，得到 I_x^2, I_y^2 ✓ (B)
3. 将它们**相加**： $I_x^2 + I_y^2$ ，而不是相乘 ✗
4. 对结果逐像素开平方： $\sqrt{I_x^2 + I_y^2}$ ✓ (D)

因此，网站上的错误步骤是 **C: Multiply the resulting images from the previous step pixelwise.**

Q16-D

Which one of the following statements about feature descriptors is incorrect?

- A. Haralick features are derived from gray-level co-occurrence matrices.
- B. SIFT achieves scale invariance by computing gradient histograms at multiple scales.
- C. LBP describes local image texture and can be multiresolution and rotation-invariant.
- D. Colour moments have higher representation capability than the colour histogram.

解析

逐一分析各选项关于特征描述子的表述：

1. **选项 A**：Haralick 特征是从 ** 灰度共生矩阵 (GLCM) ** 中提取的纹理特征（如对比度、相关性等）。**表述正确。**
2. **选项 B**：SIFT 通过构建**尺度空间（多尺度下的高斯差分金字塔）**，并在不同尺度上计算梯度直方图，从而实现尺度不变性。**表述正确。**
3. **选项 C**：LBP（局部二值模式）是描述局部纹理的特征，通过扩展可实现多分辨率（如分块 LBP）和旋转不变性（如将二进制模式旋转到最小值）。**表述正确。**
4. **选项 D**：颜色矩仅描述颜色的统计特征（均值、方差、偏度），而颜色直方图能更细致地刻画颜色的分布信息。因此颜色矩的表示能力**弱于**颜色直方图。**表述错误。**

答案

错误的选项是 **D**。

Q17-B-形状特征

Question 17

Which one of the following statements about shape feature descriptors is correct?

- A. Shape features are designed to be invariant to changes in viewpoint and illumination.
- B. Simple shape descriptors such as circularity are typically extracted after image segmentation.
- C. Shape context is computed from the histogram of gradients in a log-polar coordinate system.
- D. HOG descriptor requires segmentation to find the window for calculating gradient orientations.

解析

逐一分析各选项关于形状特征描述子的表述：

1. **选项 A**：形状特征通常不具备“视点（viewpoint）不变性”（不同视角下形状会发生几何变形）；且形状特征一般与光照无关，但“视点不变性”不是其设计目标。**表述错误。**
2. **选项 B**：圆形度（circularity）等简单形状描述子，需要先通过图像分割得到目标区域（确定形状的边界 / 区域），才能计算其形状特征。**表述正确。**
3. **选项 C**：形状上下文（Shape Context）是通过统计“某点到其他点的相对极坐标分布直方图”得到的，并非“梯度直方图”。**表述错误。**
4. **选项 D**：HOG 描述子是在**滑动窗口**中计算梯度方向直方图，不需要先进行图像分割（分割是确定目标区域，而 HOG 是基于窗口的特征提取）。**表述错误。**

答案

正确的选项是 **B**。

Q18-B-像素值

Question 18

Which one of the following statements is incorrect about why pixel values are normally not used directly as features?

- A. Pixel values may change with illumination intensity.
- B. Pixel values can be affected by camera orientation.
- C. Pixel values are typically highly redundant.
- D. Pixel values do not possess translation invariance.

解析

逐一分析各选项关于“为何不直接用像素值作为特征”的表述：

1. **选项 A**：像素值会随光照强度变化（如亮度提升会整体提高像素值），导致特征不稳定。**表述正确。**
2. **选项 B**：相机朝向（视角）改变会影响物体的成像形状 / 位置，但不会直接改变像素值本身（像素值由光照、物体反射率等决定，与视角无直接关联）。**表述错误。**
3. **选项 C**：相邻像素的像素值通常高度相似（如同一物体区域），存在大量冗余信息，不适合直接作为特征。**表述正确。**
4. **选项 D**：像素值不具备平移不变性（目标平移后，其像素值对应的位置会改变，特征会完全不同）。**表述正确。**

答案

错误的选项是 *B*。

Q19-D-灰度共生矩阵

Question 19

分享

Given the following trinary image:

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1 \end{bmatrix}$$

Which one of the following is the corresponding gray-level co-occurrence matrix $P(1, 0)$?

A.

$$\begin{bmatrix} 4 & 2 & 0 \\ 2 & 6 & 4 \\ 0 & 2 & 4 \end{bmatrix}$$

B.

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

C.

$$\begin{bmatrix} 2 & 1 & 0 \\ 1 & 3 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

D. None of the above.

解题过程（详细）

1 共生矩阵 $P(1, 0)$ 的含义

- 这是灰度共生矩阵（Gray-Level Co-occurrence Matrix, GLCM）
 - $(1, 0)$ 表示：只考虑**水平方向向右相邻**的像素对
 - 图像中灰度值为 $\{0, 1, 2\}$ ，所以 P 是一个 3×3 矩阵
 - 行：当前像素值（from）
 - 列：右侧邻居像素值（to）
- 即：

$$P(i, j) = \text{“左像素为 } i, \text{ 右像素为 } j \text{ 的像素对数量”}$$

2 列出所有水平相邻像素对

对每一行，从左到右取相邻像素对：

原图：

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 2 \\ 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 1 \end{bmatrix}$$

第一行：0 0 1 1

- $(0 \rightarrow 0)$
- $(0 \rightarrow 1)$
- $(1 \rightarrow 1)$

第二行：0 0 1 2

- $(0 \rightarrow 0)$
- $(0 \rightarrow 1)$
- $(1 \rightarrow 2)$

第三行：1 1 1 2

- $(1 \rightarrow 1)$
- $(1 \rightarrow 1)$
- $(1 \rightarrow 2)$

第四行：1 2 2 1

- $(1 \rightarrow 2)$
 - $(2 \rightarrow 2)$
 - $(2 \rightarrow 1)$
-

3 统计每种灰度对的出现次数

我们统计所有的有序对 $(i \rightarrow j)$:

- $(0 \rightarrow 0)$: 出现 2 次
- $(0 \rightarrow 1)$: 出现 2 次
- $(0 \rightarrow 2)$: 0 次
- $(1 \rightarrow 0)$: 0 次
- $(1 \rightarrow 1)$: 3 次
- $(1 \rightarrow 2)$: 3 次
- $(2 \rightarrow 0)$: 0 次
- $(2 \rightarrow 1)$: 1 次
- $(2 \rightarrow 2)$: 1 次

4 写成 3×3 共生矩阵

行/列顺序按灰度值 0,1,2 排列:

$$P(1,0) = \begin{bmatrix} P(0,0) & P(0,1) & P(0,2) \\ P(1,0) & P(1,1) & P(1,2) \\ P(2,0) & P(2,1) & P(2,2) \end{bmatrix} = \begin{bmatrix} 2 & 2 & 0 \\ 0 & 3 & 3 \\ 0 & 1 & 1 \end{bmatrix}$$

5 与选项比较

将结果与 A、B、C 三个 3×3 矩阵逐项对比，**都不相同**，因此：

正确选项为：**D. None of the above.**

如果你之后拿到别的 GLCM 题目，也可以照这个套路来：

1. 写出所有相邻像素对（按指定方向）
2. 统计每一种 $(i \rightarrow j)$ 的次数
3. 填进矩阵 $P(i, j)$ 中

Q20-D-SIFT

Question 20

Which one of the following statements about SIFT is incorrect?

- A. SIFT summarises texture in localised regions around keypoints.
- B. SIFT descriptor is designed to be invariant to scaling as well as rotation.
- C. SIFT keypoints correspond to extrema in the scale-space of the image.
- D. SIFT calculates feature vectors with lengths depending on image complexity.

解析

逐一分析各选项关于 SIFT 的表述：

1. **选项 A**：SIFT 会在关键点周围的局部区域内，通过统计梯度直方图来概括纹理信息，从而生成描述子。**表述正确**。
2. **选项 B**：SIFT 通过尺度空间极值检测实现**尺度不变性**，通过关键点主方向对齐实现**旋转不变性**，是其核心特性之一。**表述正确**。
3. **选项 C**：SIFT 的关键点是在 ** 图像的尺度空间（高斯差分金字塔） ** 中检测到的局部极值点（极大 / 极小值）。**表述正确**。
关键点就是尺度空间中 DoG 的极大 / 极小值 → 正确
4. **选项 D**：SIFT 的特征向量长度是固定的（标准 SIFT 为 128 维），与图像复杂度无关。**表述错误**。

答案

错误的选项是 *D*。

Q21-B-BoW SIFT

Question 21

Which one of the following statements about BoW feature encoding for SIFT is incorrect?

- A. BoW uses a vocabulary constructed from a training set of SIFT keypoint descriptors.
- B. BoW assigns each SIFT keypoint descriptor to the visual word with the largest distance.
- C. BoW counts the number of SIFT keypoint descriptors assigned to each visual word.
- D. BoW encodes all SIFT keypoint descriptors of an image into a fixed-dimensional histogram.

解析

逐一分析各选项关于 SIFT 的 BoW（词袋）特征编码的表述：

- 1. **选项 A**：BoW 的“视觉词汇表”是通过对训练集的 SIFT 描述子进行聚类（如 K-Means）构建的。**表述正确。**
- 2. **选项 B**：BoW 是将每个 SIFT 描述子分配给**距离最近**的视觉词（而非“距离最大”），通过最近邻匹配实现分类。**表述错误。**
- 3. **选项 C**：BoW 的核心步骤是统计每个视觉词对应的 SIFT 描述子数量，以此量化图像的特征分布。**表述正确。**
- 4. **选项 D**：BoW 会将图像的所有 SIFT 描述子编码为固定维度的直方图（维度等于视觉词汇表的大小）。**表述正确。**

答案

错误的选项是 *B*。

Q22-圆形成度

Question 22

Given an image containing one object with area and perimeter both equal to 25 pixels. What is the circularity of the object (rounded to 2 decimals)?

Options

- A. 0.04
- B. 0.25
- C. 0.50
- D. 1.00

解析

圆形成度 (Circularity) 的计算公式为: $Circularity = \frac{4\pi \times Area}{Perimeter^2}$

已知条件: 面积 (Area) =25, 周长 (Perimeter) =25, 代入公式:

$$Circularity = \frac{4 \times \pi \times 25}{25^2} = \frac{100\pi}{625} \approx \frac{314.16}{625} \approx 0.50$$

答案

正确选项是 *C*（0.50）。

Q23-模式识别-B

Question 23

Which one of the following statements about pattern recognition is correct?

- A. Pattern recognition is defined as the process of training a model on a training dataset and then testing it on an independent test set.
- B. The number of training samples should be much larger than the number of hyperparameters of the model to avoid the overfitting problem.
- C. The simple kNN classifier makes no a priori assumptions and performs well even for nonhomogeneous feature types and scales.
- D. SVM is a powerful classifier that can separate classes even when the feature space exhibits significant overlaps between classes.

题目 / Question

Which one of the following statements about pattern recognition is correct?

关于模式识别，哪个陈述是正确的？

- A) 模式识别被定义为在训练集上训练模型然后在独立测试集上测试的过程
- B) 训练样本数量应远大于模型超参数数量以避免过拟合
- C) 简单的kNN分类器不做先验假设，即使对非同质特征类型和尺度也表现良好
- D) SVM是强大的分类器，即使特征空间中类别存在显著重叠也能分离类别

答案 / Answer: **B**

解析 / Explanation

选项	正确?	说明
A	X	这只是机器学习的流程，不是模式识别的定义
B	✓	训练样本 >> 参数数量是避免过拟合的基本原则
C	X	错误！kNN对特征尺度非常敏感，需要归一化
D	X	SVM在类别严重重叠时效果也会变差

重点解析 / Key Points

为什么C是错误的：

kNN 基于距离计算，不同尺度的特征会导致问题：

特征1: [0, 1] 范围小
特征2: [0, 1000] 范围大 → 主导距离计算！

必须进行特征归一化才能让 kNN 正常工作。

为什么B是正确的：

这是机器学习防止过拟合的基本原则：

- 样本数 >> 参数数（或超参数数）
- 否则模型容易"记住"训练数据而非学习规律

C

Question: Which one of the following statements is correct for pattern recognition?

模式识别不仅仅是训练集测试集，还包括特征提取、数据预处理、模型选择等

- A. Pattern recognition is defined as the process of model training on a training dataset and then testing on an independent test set.
- B. The dimension of feature vectors should be smaller than the number of training samples in order to avoid the overfitting problem. 特征向量的维度不一定小于训练样本的数量
- C. The simple kNN classifier needs homogeneous feature types and scales so that the classification performance can be better. KNN 需要同质化的特征类型和尺度
- D. SVM is a powerful classifier that can separate classes even when the feature space exhibits significant overlaps between classes. SVM当特征空间有重叠时，它无法显著区分类别
-

Q24-半监督-C

Question 24

What is the name of the learning paradigm that uses a combination of labelled and unlabelled data to train a pattern recognition model?

Options

- A. Supervised learning
- B. Unsupervised learning
- C. Semisupervised learning
- D. Weakly supervised learning

解析

各学习范式的核心特征：

- **监督学习 (A)**：仅使用带标签数据训练模型。
- **无监督学习 (B)**：仅使用无标签数据训练模型。
- **半监督学习 (C)**：结合带标签数据和无标签数据训练模型，利用无标签数据提升模型泛化能力。
- **弱监督学习 (D)**：使用“弱标签”数据（如不精确、部分标注的标签）训练模型，并非结合标签 / 无标签数据。

答案

正确选项是 C。

Q25-D

Question 25

Which one of the following statements about nearest class mean classification is incorrect?

- A. It is one of the simplest and fastest classifiers.
- B. It works well when classes are far from each other.
- C. It performs poorly when classes are multimodal.
- D. It can handle outliers and noisy data well.

解析

近类均值分类（Nearest Class Mean, NCM）的核心是：计算每个类别的均值向量，将测试样本分配给距离最近的类均值。逐一分析选项：

1. 选项 A：

NCM 仅需计算类均值和距离，是最简单、运算最快的分类器之一。

表述正确

。

2. 选项 B：

当类别间距离较远时，类均值能清晰区分不同类别，模型效果较好。

表述正确

。

3. 选项 C：

多模态类别（同一类包含多个分布簇）的类均值会偏离真实簇中心，导致分类效果差。

表述正确

。

4. 选项 D：

NCM 的类均值易受离群点、噪声数据的影响（离群点会大幅改变类均值），无法很好处理这类数据。

表述错误

。

答案

错误选项是 *D*。

Q26-决策树-A

Question 26

Which one of the following statements about decision tree classification is incorrect?

- A. It can handle categorical but not numerical data.
- B. It is robust to outliers and missing data values.
- C. It is easy to interpret the decisions of the model.
- D. It gives information on the importance of features.

解析

逐一分析决策树分类的相关表述：

1. 选项 A：

决策树既可以处理分类数据（如类别标签），也能处理数值型数据（通过对数值特征进行划分，如“年龄>30”）。

表述错误

。

2. 选项 B：

决策树对离群点不敏感（划分规则基于局部统计），且可通过缺失值处理策略（如按多数类分配）适配缺失数据。

表述正确

。

3. 选项 C：

决策树以“树形分支”展示分类逻辑，可直观解释模型的决策过程。

表述正确

。

4. 选项 D:

决策树的特征划分顺序（或特征重要性指标）可反映特征对分类结果的贡献程度。

表述正确

。

答案

错误选项是 A。

Q27-A-随机森林

Question 27

Which one of the following statements about random forest classifiers is correct?

- A. Increasing the correlation between the individual trees increases the random forest classification error rate.
- B. Increasing the number of selected features at each node decreases the correlation between the individual trees.
- C. Increasing the number of selected features at each node decreases the strength of the individual trees.
- D. Increasing the strength of the individual trees increases the random forest classification error rate.

解析

随机森林的核心是“多棵决策树的集成”，依赖**树之间的低相关性**和**单棵树的强分类能力**来提升性能，逐一分析选项：

1. 选项 A:

树之间的相关性越高，集成的多样性越低，随机森林的分类误差会增加

（而非“错误表述”）。但需结合其他选项验证。

2. 选项 B:

每个节点选择的特征数越多，树的构建逻辑越相似，树之间的相关性会增加

（而非“降低”）。

表述错误

。

3. 选项 C:

每个节点选择的特征数越多，单棵树能利用更丰富的信息，分类能力（强度）会增强

（而非“降低”）。

表述错误

。

4. 选项 D:

单棵树的强度（分类能力）越强，集成后的随机森林误差会降低

（而非“增加”）。

表述错误

。

修正对选项 A 的分析：随机森林的误差由“树的强度”和“树的相关性”共同决定——**树之间的相关性越高，集成的泛化能力越差，误差率会增加**。因此选项 A 的表述是正确的。

答案

正确选项是 **A**。

增加 correlation => 增加 error rate

减少 correlation => 减少 error rate => 增加泛化能力

增加 m 选择的特征数量 => 增加 correlation & strength

减少 m 选择的特征数量 => 减少 correlation & strength

越相关，森林中的树很像，误差率越大

Q28-A-SVM

Question 28

Which one of the following statements about SVM classifiers is incorrect?

- A. Standard SVMs aim to minimize the margin around the decision boundary.
- B. Hard-margin SVMs assume that the classes are entirely linearly separable.
- C. Soft-margin SVMs use slack variables to allow for misclassification of samples.
- D. Nonlinear SVMs map samples into a higher-dimensional space using a kernel function.

解析

逐一分析 SVM 分类器的表述：

1. 选项 A：

标准 SVM 的核心目标是

最大化

决策边界的间隔（Margin），间隔越大模型泛化能力越强，并非“最小化”。

表述错误

。

2. 选项 B：

硬间隔 SVM 的前提是“类别完全线性可分”，此时能找到无错分样本的最优间隔超平面。

表述正确

。

3. 选项 C：

软间隔 SVM 引入松弛变量，允许部分样本错分，以适配类别不完全可分的场景。

表述正确

。

4. 选项 D：

非线性 SVM 通过核函数将低维非线性数据映射到高维空间，使其在高维中线性可分。

表述正确

。

答案

错误选项是 *A*。

Q29-C-recall

Question 29

Which one of the following statements about segmentation evaluation metrics is incorrect?

- A. Sensitivity is the fraction of the true object that is correctly segmented.
- B. Specificity is the fraction of the true background that is correctly segmented.
- C. Precision is the fraction of the true object that is correctly segmented.
- D. Recall is the fraction of the true object that is correctly segmented.

析

分割评估指标的定义：

1. **选项 A**：灵敏度（Sensitivity）即召回率（Recall），是“真实目标中被正确分割的比例”。**表述正确。**
2. **选项 B**：特异性（Specificity）是“真实背景中被正确分割的比例”。**表述正确。**
3. **选项 C**：精确率（Precision）是“分割结果中属于真实目标的比例”，而非“真实目标中被正确分割的比例”。**表述错误。**
4. **选项 D**：召回率（Recall）与灵敏度一致，是“真实目标中被正确分割的比例”。**表述正确。**

答案

错误选项是 *C*。

Q30-D

Question 30

Which one of the following statements about image segmentation methods is **incorrect**?

- A. Thresholding pixel intensities yields poor results if the regions to be segmented have overlapping intensity distributions.
- B. Level-set segmentation uses an object boundary representation that can easily deal with topological changes.
- C. Region merging techniques aim to optimize some metric of intra-region similarity and/or inter-region dissimilarity.
- D. Watershed segmentation often results in under-segmentation because image regions may contain multiple local minima.

解析

题干问的是：

Which statement is **incorrect**?

逐项分析：

选项	判定	原因
A Thresholding pixel intensities yields poor results if regions have overlapping intensity distributions.	✓ 正确	阈值分割依赖像素强度不同，若分布重叠则难分割
B Level-set segmentation handles topological changes.	✓ 正确	Level-set 可自然处理分裂 / 合并的边界形状变化
C Region merging aims to optimize intra-region similarity & inter-region dissimilarity.	✓ 正确	区域增长与合并追求区域内部一致且区域间差异明显
D Watershed segmentation often results in under-segmentation because image regions may contain multiple local minima.	✗ 错误！	Watershed 的典型问题是 过分割 (over-segmentation) ，不是欠分割

✦ 为什么 D 错？

Watershed 基于灰度拓扑结构：

- 每个局部极小点 == 一个初始“盆地”
- 图像中噪声与纹理会产生大量局部极小点
- 这会导致：

严重过分割 (over-segmentation)

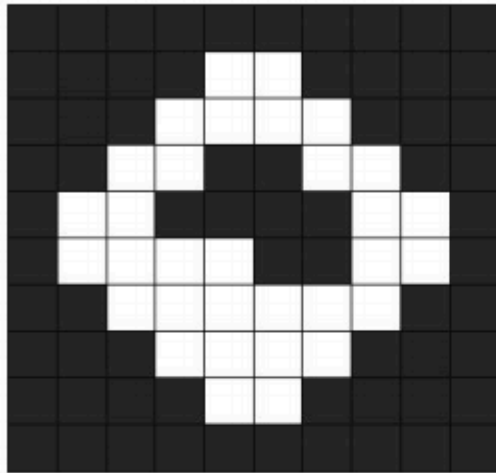
因此 D 的说法与常识相反，故为错误陈述。

Q31-膨胀-A

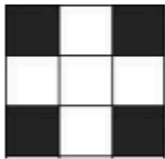
Question 31

B

Given the following binary image / (where white is object and black is background):

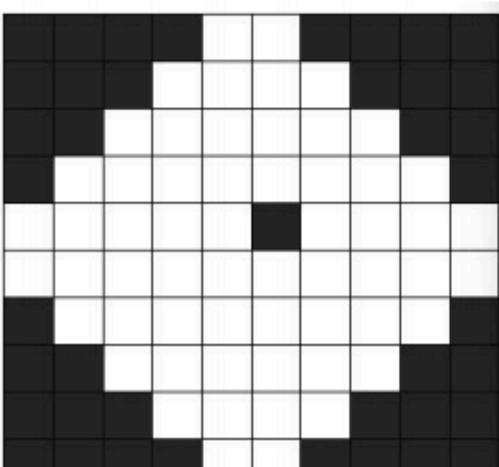


and the following structuring element S (the centre pixel being the origin):

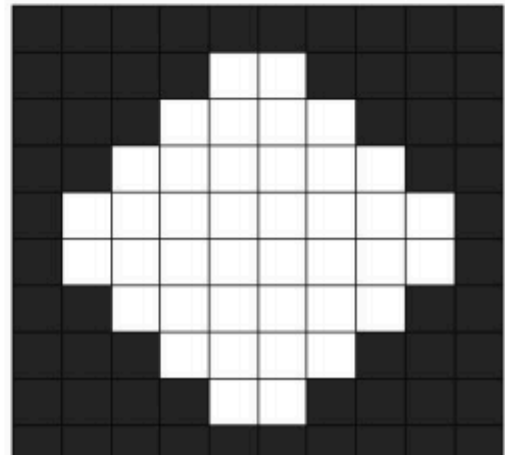


Which one of the following images is equals to $I \oplus S$?

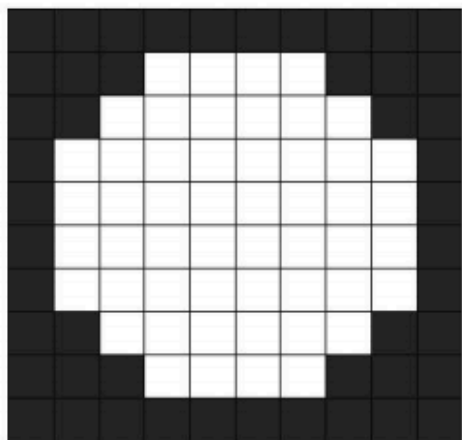
A.



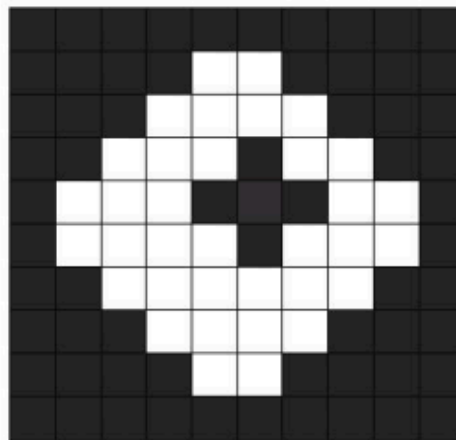
B.



C.



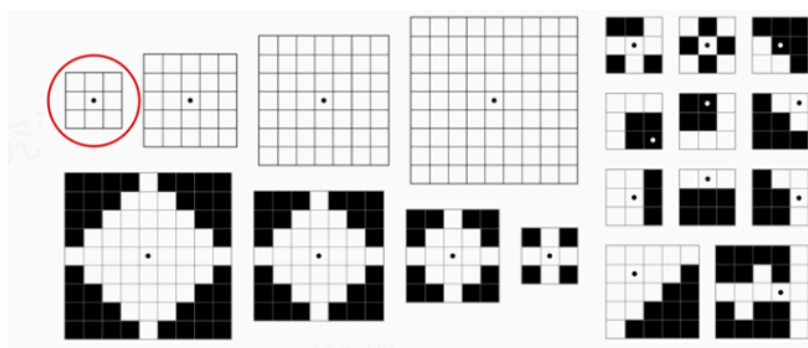
D.



结构元素

结构元素可以有任意形状，结构元素的形状对膨胀和腐蚀的结果有显著影响：

- 对称的结构元素可以均匀地处理图像边缘。
- 不对称或不同形状的结构元素可以用来处理特定方向或形状的细节。



结构元素 S 是一个以中心像素为原点的“十字”形（中心及其上下左右为 1）。

膨胀 $I \oplus S$ ：把这个十字形在所有前景像素上“扫一遍”，结果就是把物体向上下左右扩展 1 像素。

对于原来的“空心菱形环”，膨胀后：

- 外边界向外扩一圈 → 外轮廓略变粗；
- 内边界向内扩一圈 → 中间空洞变小，并最终只剩下一个黑色的孔洞。

Q32-B-ISSR

Question 32

Given a binary input image I to which we apply the following algorithm:

1. "Compute the complement C of input image I ."
2. "Copy the border pixels of C into new image S ."
3. "Compute the reconstruction R of C from S ."
4. "Compute the complement image O of R ."

What does the output image O contain?"

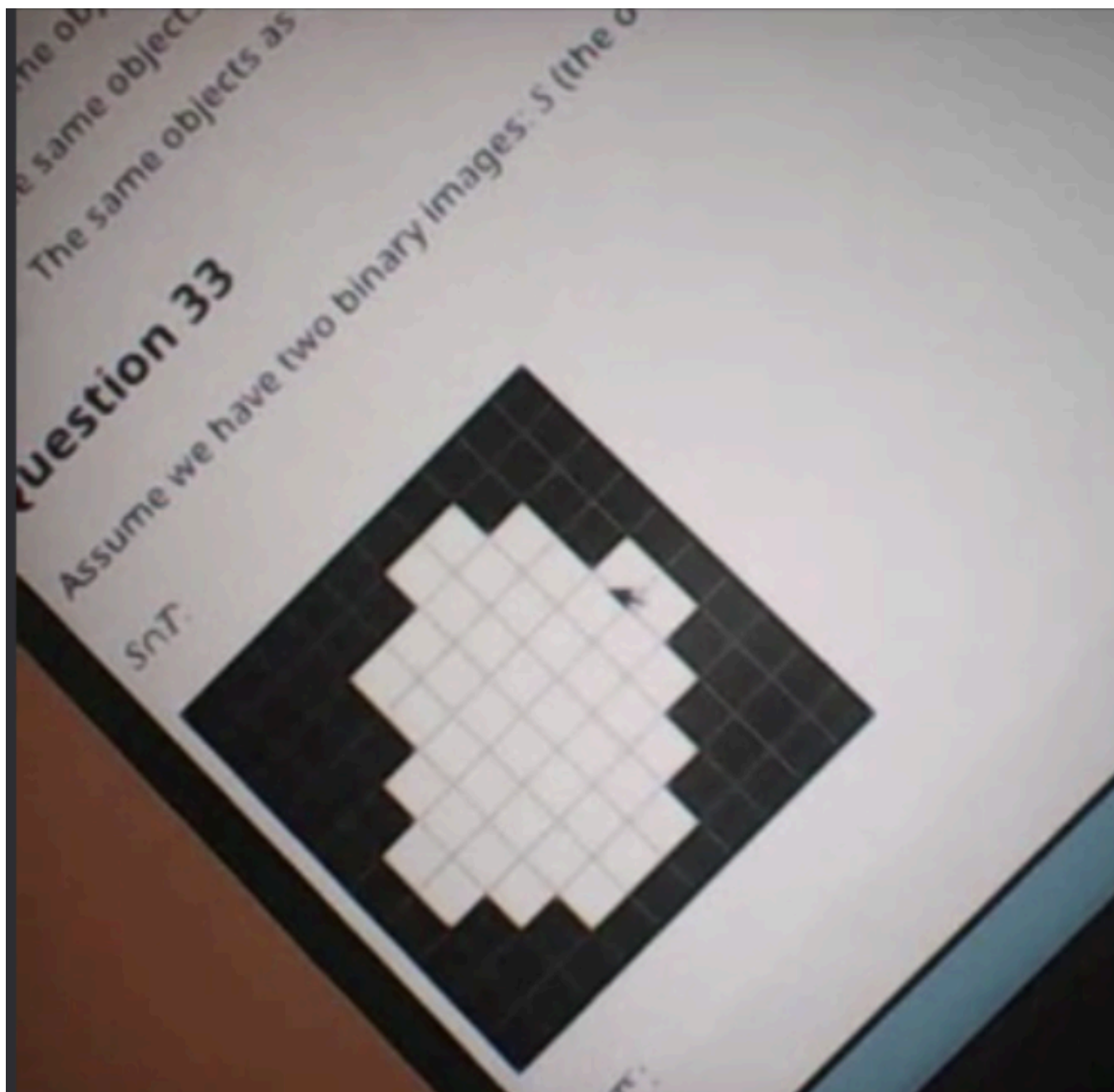
选项 (Options): (根据图片碎片还原的选项)

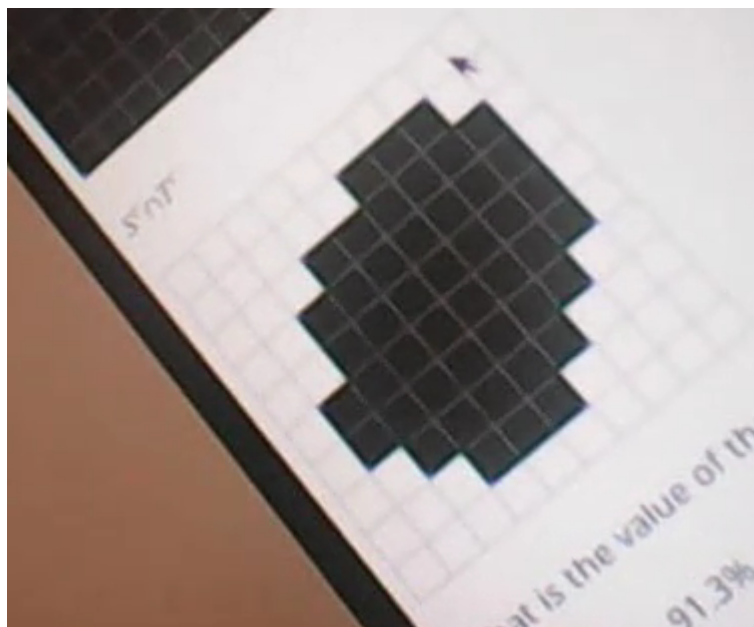
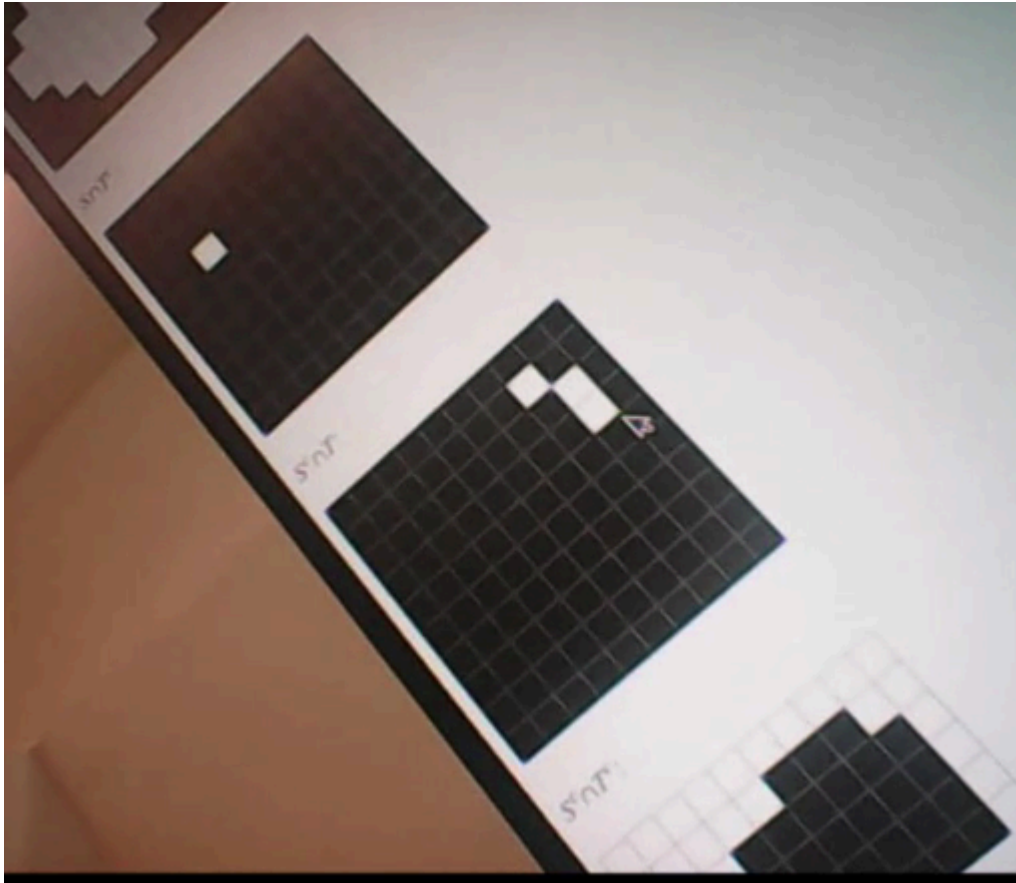
- **A.** The same objects as the input image whatever the input.
- **B. The same objects as the input image but with holes filled.** (与输入图像相同的物体，但孔洞被填充了。)
- **C.** The same objects as the input image except the boundary objects. (去除边界物体后的输入图像。)
- **D.** The same objects as the input image but with touching objects separated. (将粘连物体分离后的图像。)

选项大意为：

- A. 与输入图像包含相同的对象（无论输入是什么）；
- B. 与输入图像包含相同的对象，但内部的孔洞被填满；
- C. 与输入图像相同，但去掉所有与边界相连的对象；
- D. 与输入图像相同，但把互相接触的对象分开。

Q33-Jaccard 相似系数(放弃)

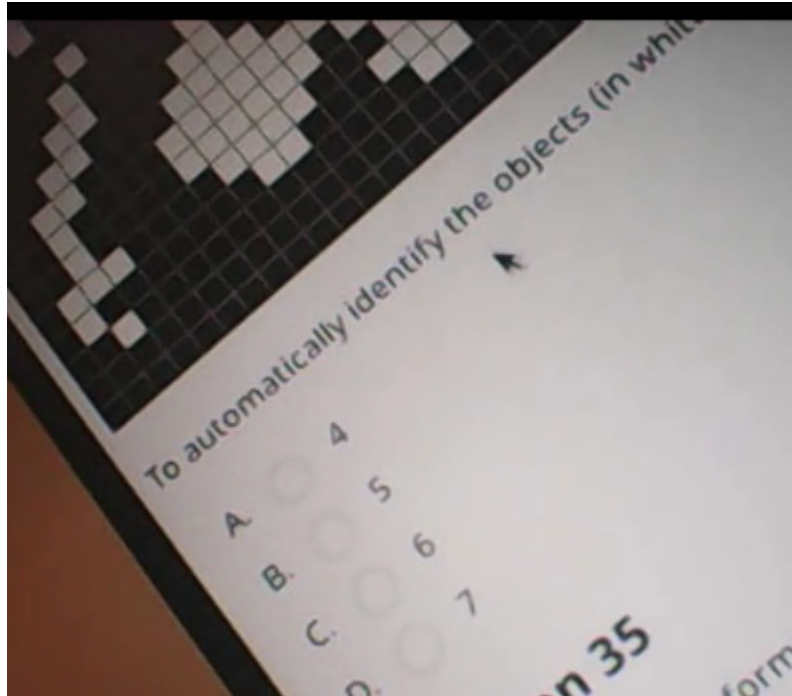




Assume we have two binary images **S** (the output of a segmentation method) and **T** (the true segmentation made manually by an expert).

The images shown give all possible intersections of these two images and/or their complements:

- $S \cap T$ – pixels that are object in both S and T (true positives, white的大块区域)
- $S \cap \bar{T}$ – pixels that are object only in S (false positives, 只有 1 个白格)
- $\bar{S} \cap T$ – pixels that are object only in T (false negatives, 有 2 个白格)



Given the following binary image after segmentation (white = object, black = background).

To automatically identify the objects (in white) in this image we can use the connected components labelling algorithm.

How many separate objects will this algorithm find here if it uses 8-connectivity?

选项（从图上看）：

Q35-D

Question 35

题干重构

To compare the performance of segmentation methods, researchers often use receiver operating characteristic (ROC) analysis.

Which one of the following statements about the ROC plot is incorrect?

- A. The ROC plot shows the relation between the sensitivity and specificity.
- B. The ROC plot of a method performing random pixel labelling is a straight line.
- C. The ROC plot of a better method lies above the ROC plot of a worse method.
- D. The ROC plot is often quantitatively summarized by the area above the curve.

解析

ROC 曲线的核心定义与特性：

1. **选项 A**：ROC 曲线以“灵敏度（Sensitivity）”为纵轴、“1 - 特异性（1-Specificity）”为横轴，本质是展示灵敏度与特异性的关系。**表述正确。**
2. **选项 B**：随机像素标注的方法，其灵敏度与 1 - 特异性呈线性关系，ROC 曲线为从 (0,0) 到 (1,1) 的直线。**表述正确。**
3. **选项 C**：性能更优的方法，在相同 1 - 特异性下灵敏度更高，因此其 ROC 曲线位于性能较差方法的曲线上方。**表述正确。**
4. **选项 D**：ROC 曲线的定量总结指标是“曲线下面积（AUC）”，而非“曲线上方面积”。**表述错误。**

答案

错误选项是 *D*。

D的话，把above改成under就对了

Q36-C

Question 36

题干（合并）：

Given a binary input image I containing various segmented objects.

Let S be the 8-connected structuring element.

What does the output image $(I \oplus S) - (I \ominus S)$ contain?

- A. The outer edges of the objects.
- B. The inner edges of the objects.
- C. The outer plus the inner edges of the objects.
- D. The outer minus the inner edges of the objects.

解析：

- $I \oplus S$ ：把物体向外“长一圈”——得到比原来更大的区域（外扩一圈）。
- $I \ominus S$ ：把物体向内“缩一圈”——得到比原来更小的区域（内缩一圈）。
- $(I \oplus S) - (I \ominus S)$ ：取“外扩结果减去内缩结果”，中间那一圈带状区域被保留下来，刚好覆盖从内边界到外边界的整个边缘带，也就是“内外边界一起”。

因此包含的是外边缘 + 内边缘。

答案：C. The outer plus the inner edges of the objects.

Q37-B

Question 37

题干（合并）：

Given a grayscale input image I and a flat structuring element S having a circular shape of a certain radius.

What does the output image $I \bullet S = (I \oplus S) \ominus S$ contain?

- A. An image in which all pixels are zero regardless of the input image I .
- B. An image in which the dark spots of I smaller than S are removed.
- C. An image in which bright spots of I smaller than S are removed.
- D. An image in which all spots of I smaller than S are removed.

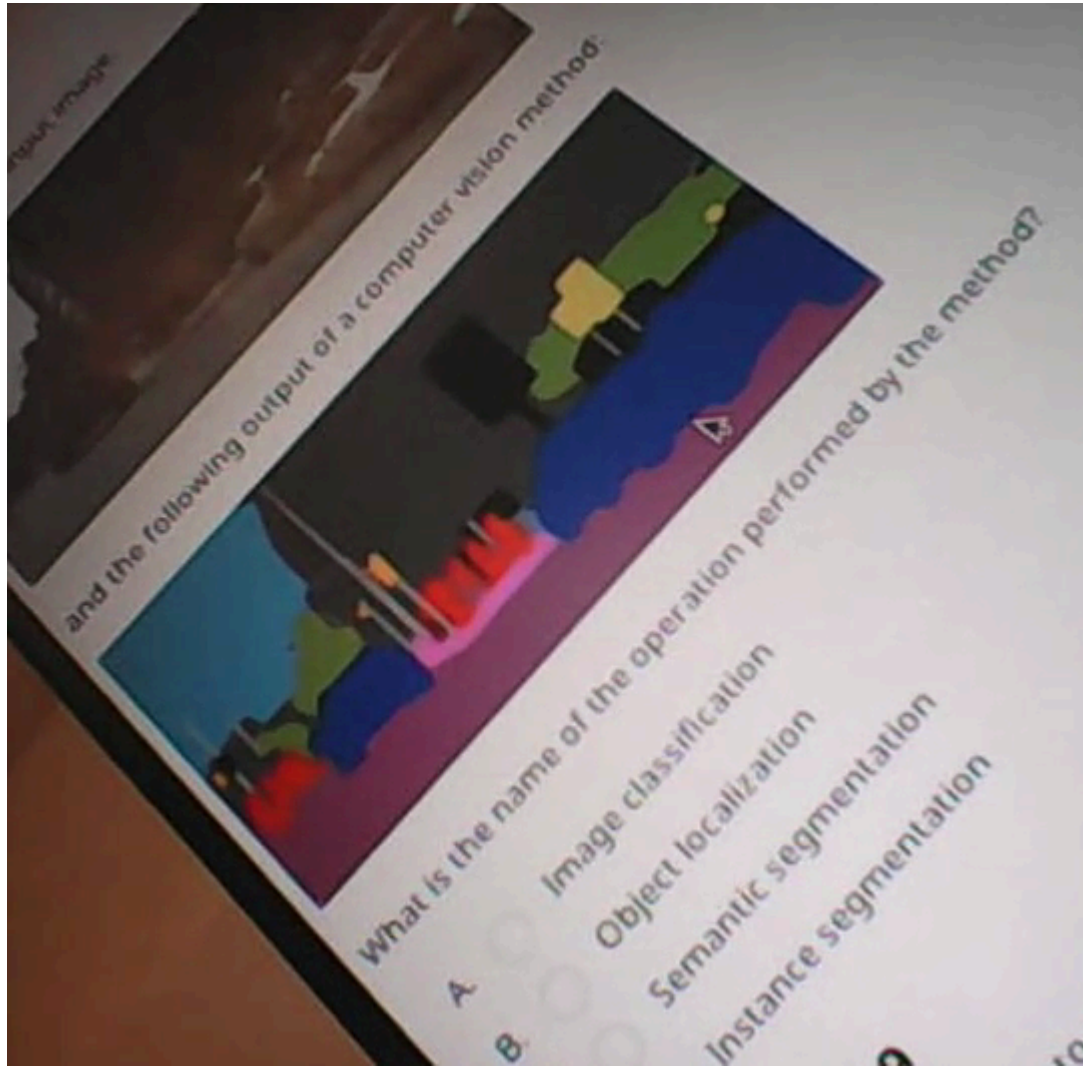
解析：

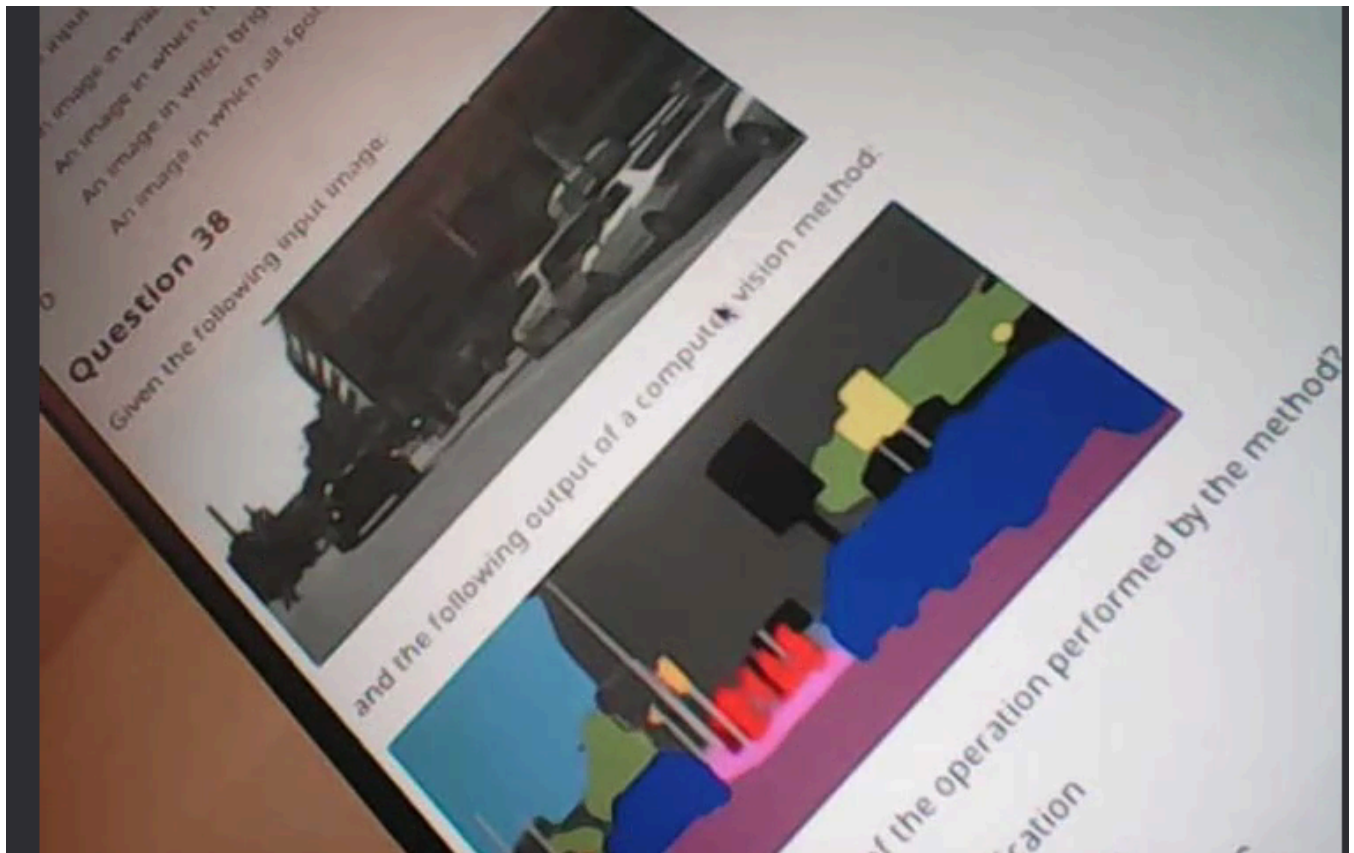
- $(I \oplus S) \ominus S$ 是灰度形态学的 closing（闭运算）。
- 闭运算的主要效果：
 - 保留并稍微平滑亮区域的轮廓；
 - 填平/去掉小的暗孔洞和窄缝（即比结构元素 S 小的“暗斑”）。
- 不会把小的亮斑去掉，那是 opening（开运算， $(I \ominus S) \oplus S$ ）的效果。

所以正确描述是：去掉比结构元素小的暗斑。

答案：B. An image in which the dark spots of I smaller than S are removed.

Q38-C





Question 38

题干 (英文)

Given the following input image and the following output of a computer vision method, what is the name of the operation performed by the method?

题干 (中文)

给定如下输入图像以及某计算机视觉方法得到的输出结果，这个方法执行的操作名称是什么？

选项

- A. Image classification (图像分类)
- B. Object localization (目标定位)
- C. Semantic segmentation (语义分割)
- D. Instance segmentation (实例分割)

正确答案: C. Semantic segmentation 语义分割

理由:

输出图像中，每个像素被赋予一个语义类别（天空、建筑、道路、草地等），整幅图被按“语义区域”用不同颜色分块显示，而**没有区分同类中的不同实例**（例如路上若有多辆车，并没有给每一辆车一个单独的 id）。这正是语义分割的典型输出形式，而不是只给整幅图一个标签（分类），也不是标框（定位或目标检测），也不是对同类不同实例分别标注（实例分割）。

Q39-A?

Question 39

One possible approach to segmenting images into meaningful regions is to transform all pixels in the image space to points in the feature space, then apply the mean-shift algorithm in the feature space to find clusters of points, and finally give each pixel the label of its corresponding cluster. Which one of the following statements about this segmentation method is **incorrect**?

- A. The method does not require specifying the number of regions.
- B. The method scales well with the dimensionality of the feature space.
- C. The method is very robust to outliers in the feature space.
- D. The method has only one parameter: the window size.

Question 39

题干 (英文)

One possible approach to segmenting images into meaningful regions is to transform all pixels in the image space to points in the feature space, then apply the mean-shift algorithm in the feature space to find clusters of points, and finally give each pixel the label of its corresponding cluster. Which one of the following statements about this segmentation method is **incorrect**?

题干 (中文)

一种将图像分割成有意义区域的方法是：把图像空间中的所有像素映射到特征空间中的点，然后在特征空间中对这些点应用 **mean-shift (均值漂移)** 算法寻找聚类，最后把每个像素赋予其所属聚类的标签。关于这种分割方法，下列哪一项说法是 **错误** 的？

选项 (按屏幕内容还原) |

- A. The method does not require specifying the number of regions.
(该方法不需要事先指定区域/聚类的个数。)
- B. The method scales well with the dimensionality of the feature space.
(该方法在特征空间维度升高时仍具有良好的扩展性。)
- C. The method is very robust to outliers in the feature space.
(该方法对特征空间中的离群点非常鲁棒。)
- D. The method has only one parameter, the window size.
(该方法只有一个参数，即窗口大小 / 核带宽。)

✓ **正确答案：B?**

■ 解析

题目描述的方法是 **Mean-Shift Segmentation (均值漂移分割)**

它的主要特点：

选项	是否正确	原因说明
A The method does not require specifying the number of regions.	✓	Mean-shift 自动发现聚类数目 ，无需指定 K
B The method scales well with the dimensionality of the feature space.	✗ 错误陈述	Mean-shift计算成本随着维度急剧上升（高维数据非常慢）→“维度诅咒”
C The method is very robust to outliers	✓	核密度估计使得离群点影响较低
D The method has only one parameter: window size (bandwidth)	✓	bandwidth 是核心参数，决定聚类尺度

💡 为什么 B 是错误的？

Mean-shift 复杂度高：

$$O(n^2d)$$

其中 n =数据点数， d =特征维度
→ 当特征维度大时，计算非常慢，不适合高维特征空间

因此：

Mean-shift **不适合高维数据**，因此 B 是错误的

🧠 总结记忆

Mean-shift：不设聚类数、抗噪声，但**维度高就会很慢**

Q40-B

Question 40

题干: Which one of the following best describes the purpose of the max-pooling operation in a CNN?

选项 (从图中可读) :

- A. Reducing the number of feature maps.
- B. Reducing the size of the input images.
- C. Reducing the number of model parameters.
- D. Reducing the number of network layers.

Q41-C

Question 41

题干: Which one of the following is **not** the purpose of convolutional layers in CNNs?

选项:

- A. Reducing the number of parameters.
- B. Convolving the images with kernels.
- C. Applying a nonlinear activation function.
- D. Making the network learn image features.

✅ 答案: C

理由: 非线性激活通常是独立的激活层/操作 (如 ReLU), 不是“卷积层本身”的核心目的。其余三项都是卷积层的典型作用或直接行为。

Q42-C

Question 42

题干: Which one of the following best describes the goal of transfer learning?

选项:

- A. Augmenting image data to facilitate the learning process.
- B. Training a new network model from scratch for each task.
- C. Using a pretrained model and further training it for a new task.
- D. Combining multiple networks in a single model to improve learning.

✅ **答案: C**

理由: 迁移学习就是**利用预训练模型的知识**，再对新任务进行微调/再训练。

Q43-B

Question 43

题干: Which one of the following best describes the purpose of using a stride larger than 1 in a CNN?

选项:

- A. Decreasing the size of the filter kernels.
- B. Decreasing the size of the feature maps.
- C. Decreasing the size of the input image.
- D. Decreasing the size of the output image.

✅ **答案: B**

理由: **stride > 1** 会下采样输出特征图，直接导致**feature maps** 的空间尺寸变小。

(D 也有点像，但 B 更符合术语与考点表述。)

Q44-B

Question 44

题干: Which one of the following strategies will **not** help to reduce overfitting of neural networks?

选项:

- A. Applying dropout.
- B. Adding more layers.
- C. Using early stopping.
- D. Augmenting the data.

✅ 答案: B

理由: 增加层数通常**提升模型容量**, 更可能加剧过拟合; A/C/D 都是常见正则/抑制过拟合手段。

Q45-A

Question 45

题干: Which one of the following statements about skip connections in the U-Net architecture is **incorrect**?

选项 (从图中可读) :

- A. Skip connections allow outputs from later network layers to feed back into earlier layers.
- B. Skip connections help recover fine-grained details when going from lower to higher layers.
- C. Skip connections reduce the vanishing gradient problem in the network training process.
- D. Skip connections enable feature reusability and stabilize training and convergence.

解析

U-Net 中跳跃连接 (Skip Connections) 的核心作用是**将编码器 (下采样) 阶段的特征图直接传递到解码器 (上采样) 阶段**, 逐一分析选项:

1. 选项 A:

跳跃连接是“编码器的早期层特征传递到解码器的后期层”, 而非“后期层输出反馈到早期层”(后者是循环连接的逻辑)。

表述错误

。

2. 选项 B:

编码器早期层的特征包含细粒度细节, 跳跃连接可将这些细节传递到解码器, 帮助恢复分割结果的精细信息。

表述正确

。

3. 选项 C:

跳跃连接提供了梯度传播的“捷径”，减少了梯度在深层网络中逐渐消失的问题。

表述正确

。

4. 选项 D:

跳跃连接实现了特征复用（编码器特征直接用于解码器），同时稳定了训练过程，加快收敛。

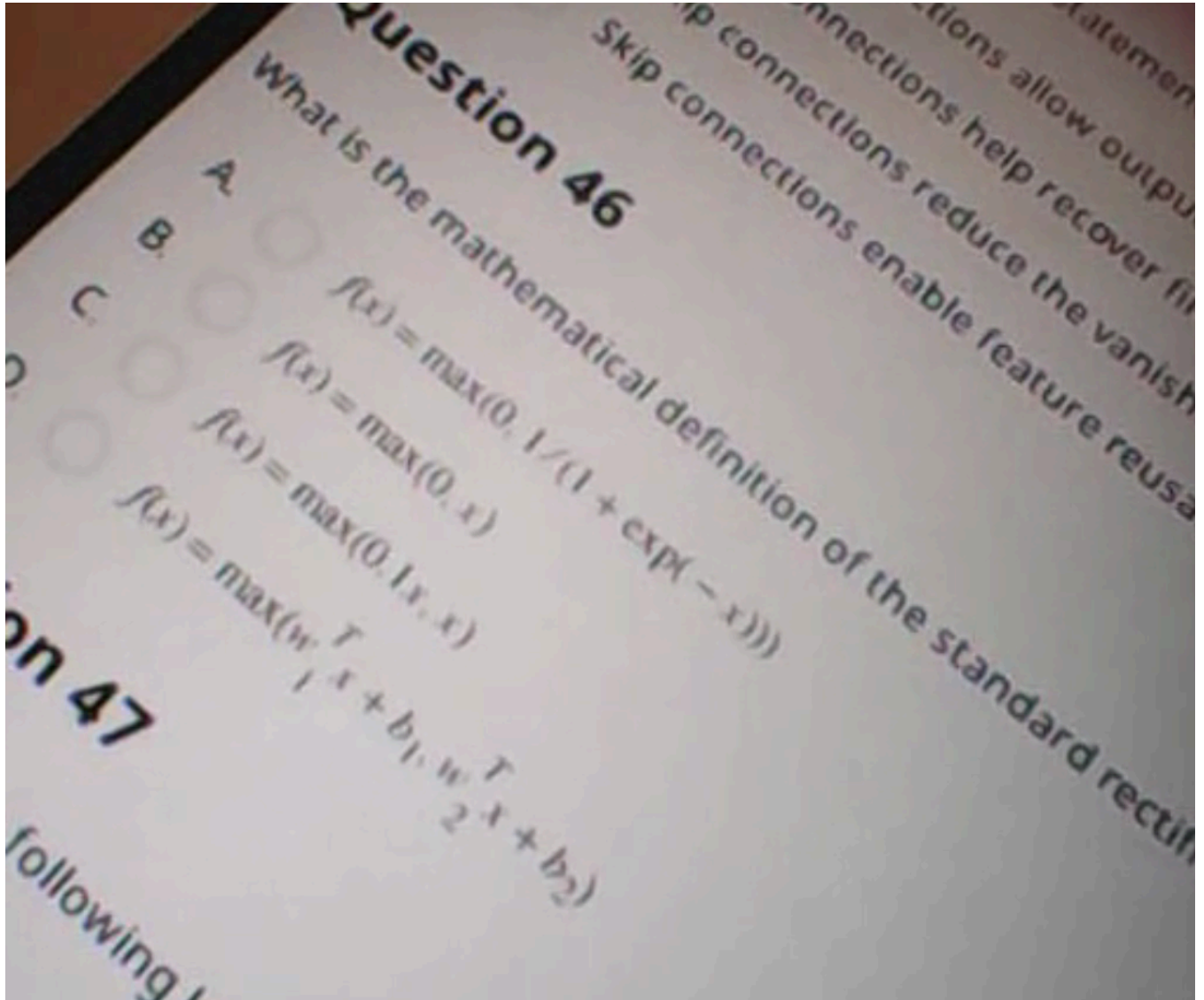
表述正确

。

答案

错误选项是 A。

Q46-B-标准RELU



Q47-C

Question 47

题干: Which one of the following kinds of neural network is most suited for object detection tasks?

选项:

- A. Multilayer perceptron (MLP)
- B. Fully convolutional network (FCN)
- C. Region proposal network (RPN)
- D. Recurrent neural network (RNN)

✅ **答案: C**

理由: RPN 是 Faster R-CNN 等检测框架的关键模块，用于生成候选框，最贴合“object detection”这个任务定位。

Q48-A?

题干: Which one of the following statements about loss functions in model training is incorrect?

- A. Loss functions compute the **prediction accuracy** of a model from its outputs.
- B. Loss functions inform the **backpropagation** process in training a model.
- C. Loss functions are commonly optimized using **gradient descent** methods.
- D. Loss functions can be used to monitor **overfitting** of a model on the test data

Q49-D

Which one of the following statements about motion analysis is **incorrect**?

- A. Detection of moving objects by subtraction of successive images in a video works best if the background is constant.
- B. Sparse motion estimation in a video can be done by template matching and minimising the sum of squared differences.
- C. Dense motion estimation using optical flow assumes that each small neighbourhood remains constant over time.
- D. Optical flow provides an equation for each pixel that can be uniquely solved independently from neighbouring pixels.

解析

逐一分析运动分析的表述：

1. **选项 A**：帧差法检测运动目标的前提是**背景恒定**，此时帧间差异主要由运动目标引起。**表述正确**。
2. **选项 B**：稀疏运动估计可通过模板匹配（对特征点周围模板计算平方差和并最小化）实现，对应光流中的稀疏方法。**表述正确**。
3. **选项 C**：稠密光流的核心假设之一是**局部亮度恒定**（小邻域内像素亮度随时间不变），以此建立光流方程。**表述正确**。
4. **选项 D**：光流的基本方程是“一个方程对应两个未知量（x、y 方向的流速）”，是**欠定问题**，无法独立求解单个像素的光流，必须结合邻域约束（如平滑性假设）。**表述错误**。

答案

错误选项是 **D**。

Q50-A?

题干： Which one of the following statements about object tracking is incorrect?

- A. The prediction step of Bayesian inference assumes that the next object state can be calculated using only the posterior of the previous state.
- B. The hidden Markov model framework assumes that the current state of the object to be tracked depends only on the previous state.
- C. The Kalman filtering method assumes that the probability distributions to be estimated can be accurately represented by Gaussians.
- D. The particle filtering method assumes that the measurement model cannot be easily parameterized.

SAMPLE

Q0

Question: What is the effect of the following 2D convolution kernel on an image?

0	1	0
1	-4	1
0	1	0

- A. It approximates the sum of first-order derivatives in x and y .
- B. It approximates the sum of second-order derivatives in x and y .
- C. It approximates the product of first-order derivatives in x and y .
- D. It approximates the product of second-order derivatives in x and y .

要解析这道关于卷积核作用的题目，需从**导数的离散近似**和核的结构入手：

步骤 1：回顾导数的离散近似

- **一阶导数（梯度）**：通常用相邻像素的差值近似（如 $[1, -1]$ 核）。
- **二阶导数（拉普拉斯）**：是一阶导数的导数，离散形式为“中心像素减去周围像素的加权和”（核心是**中心系数为负，周围系数为正，且核元素和为 0**）。

步骤 2：分析该卷积核的结构

题目中的 3×3 核为：

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

这是**二维拉普拉斯核**：

- 对 x 方向二阶导数：离散近似为“当前像素 - 左右像素”，对应核 $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix}$ ；
- 对 y 方向二阶导数：离散近似为“当前像素 - 上下像素”，对应核 $\begin{bmatrix} 0 & 0 & 0 \\ 1 & -2 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ ；
- 该核是 x 、 y 方向二阶导数核的**和**（ $-2 + (-2) = -4$ ，周围系数相加为 $1 + 1 + 1 + 1 = 4$ ）。

步骤 3：分析选项

- 选项 A：一阶导数的和是梯度的模（或分量和），对应核是一阶差分核（如 $[1, -1]$ ），与该核不符。
- 选项 B：该核是x、y方向二阶导数（拉普拉斯）的和，符合分析。
- 选项 C、D：“乘积”与核的线性组合（和）性质不符。

因此，正确答案是 *B*。

Q1

Question: Given the following 2D convolution kernel:

1	0	-1
2	0	-2
1	0	-1

Which one of the following correctly describes the effect of this kernel when applied to an image?

-
- A. Smoothing in x and first-order differentiation in y .
 - B. Smoothing in x and second-order differentiation in y .
 - C. First-order differentiation in x and smoothing in y .
 - D. Second-order differentiation in x and smoothing in y .

Q2-移除接触到边界的物品

Question: Given a binary input image I to which we apply the following algorithm:

Step 1: Create a copy C of input image I .

Step 2: Copy the boundary pixels of C into new image B .

Step 3: Compute the reconstruction R of C from B .

Step 4: Compute output O by subtracting R from I .

What does the output image O contain?

- A. The same objects as the input image whatever the input.
- B. The same objects as the input image but with holes filled.
- C. The same objects as the input image except the boundary objects.
- D. The same objects as the input image but with touching objects separated.

解析

该算法是基于**形态学重建的边界对象移除**操作，步骤逻辑如下：

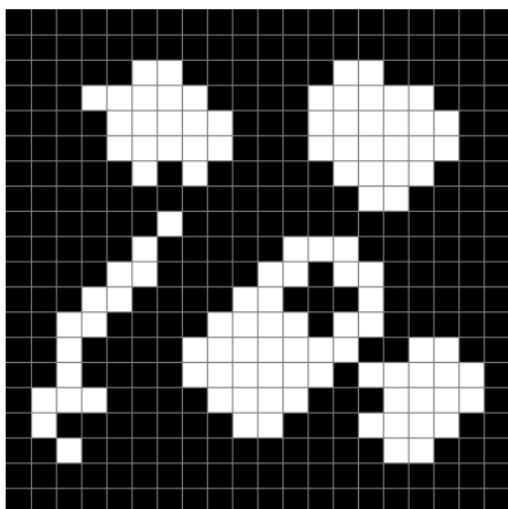
1. 复制二值图 I 得到 C ： C 与 I 完全一致；
2. 提取 C 的边界像素作为种子 B ：边界像素对应与图像边缘相连的对象区域；
3. 从 B 重建 C 得到 R ：形态学重建会填充所有与边界连通的区域（即边界对象）；
4. 用 I 减去 R 得到 O ：移除了 I 中与边界连通的对象，保留内部对象。

答案

正确选项是 C （与输入图像包含相同的对象，但去除了与边界相连的对象）。

Q3-D-四连通

D Question: Given the following binary image after segmentation.



To automatically identify the objects (in white) in this image we can use the connected components labelling algorithm. How many separate objects will this algorithm find here if it uses 4-connectivity?

- A. 4
- B. 5
- C. 6
- D. 7

Q4-B

B Question: Which one of the following statements about intensity transformations is incorrect?

- A. Contrast stretching linearly maps intensities between two values to the full output range.
- B. Log transformation maps a narrow range of high intensities to a wider range of output values.
- C. Power transformation can map intensities similar to log and inverse log transformations.
- D. Piecewise linear transformations can achieve contrast stretching and intensity slicing.

解析

逐一分析亮度变换的表述：

1. **选项 A**：对比度拉伸是将指定区间内的亮度值线性映射到全输出范围，以此增强对比度。**表述正确。**
2. **选项 B**：对数变换的特性是将窄范围的低亮度值映射到宽范围的输出值（而非高亮度），常用于增强暗部细节。**表述错误。**
3. **选项 C**：幂次变换（如伽马变换）通过调整幂次，可实现类似对数变换（幂次 < 1 ）或逆对数变换（幂次 > 1 ）的效果。**表述正确。**
4. **选项 D**：分段线性变换可通过不同线段的设计，实现对比度拉伸（扩大亮度区间）或亮度切片（提取特定亮度范围）。**表述正确。**

答案

错误选项是 *B*。